Observationally Equivalent Algorithms: What Prices Reveal About Trader Populations, and What Supervisors Need

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Abstract

Algorithmic traders read prices and, by trading, move them. One object, the feedback kernel, is a minimal sufficient statistic for such a market: for any insider strategy and causal dealer rule, trader populations with the same kernel produce the same prices, order flows, profits and equilibria, and prices identify the kernel and nothing finer. Learning the composition of the population costs data of order $(\Delta\eta)^{-(4K-2)}$ for K groups whose memories lie within $\Delta\eta$, by probing or by waiting, and the barrier persists as the market approaches instability. At illustrative parameters, two-group populations whose memories all have half-lives between 5 and 9 days need more than 4,000 years of daily prices to be told apart. Market-wide supervision therefore needs only the kernel; targeted supervision needs flow tagged by class. Core results are machine-checked in Lean 4.

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Supplement. The online supplement states every result in full, with proofs and machine-check details.

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Code and proofs. https://github.com/Pratikn03/Fianince.

1 Introduction

Trend-following funds, execution algorithms and AI trading agents read recent prices, decide what to trade and, by trading, move the prices they read. Regulators are now deciding how far to identify them. China's securities regulator issued rules on program trading in 2024, and since July 2025 its exchanges classify accounts as high-frequency by their order rates.[1] India requires every algorithmic order to carry an exchange-issued identifier.[2] Japan has registered high-speed traders since 2018, and its regulator studies their orders by trader and strategy.[3] The EU requires firms to identify their algorithms in order records under MiFID II. In the United States, the SEC withdrew its proposal on firms' use of predictive data analytics in June 2025. Behind these choices is a question this note answers: what does a market's price record reveal about its population of feedback traders, and for which decisions is that enough?

The answer. Prices reveal the market's *feedback kernel*, the response, m periods later, of total feedback flow to a unit move in the price signal. The kernel is all that prices identify, and it is all that anyone trading on public information needs. The composition of the trader population matters only for actions aimed at one group, and learning it from prices is very expensive, even on the eve of a crash: at illustrative parameters, more than 4,000 years of daily prices when the traders' memories have half-lives between 5 and 9 days (Table 1). This leaves a supervisor with a problem that the note states precisely at the end: keeping a market safe without knowing who trades in it.

Results.

(1) *Sufficiency* (Theorem 3.1). With a value, noise traders, an insider and a dealer, each following an arbitrary causal rule, two populations with the same kernel give the same prices, order flows, insider profits and dealer inventories, realization by realization. The two games therefore have the same equilibria.

(2) *Identification* (Theorem 3.2). Two populations give identical price responses to every trade sequence exactly when their *merged measures*, the trading weight at each memory rate, coincide. Groups with the same memory cannot be separated however long the record. For any population, linear or not, prices identify exactly its aggregate flow map, and the cost bounds below hold over any class that contains the linear rules.

[1] Shanghai, Shenzhen and Beijing stock exchanges, implementing rules of 3 April 2025 for the CSRC Provisions on Program Trading of May 2024.

[2] SEBI circular SEBI/HO/MIRSD/MIRSD-PoD/P/CIR/2025/0000013, 4 February 2025.

[3] Financial Services Agency, FSA Analytical Notes, 21 November 2024.

(3) *Cost* (Theorem 4.1, Proposition 4.2). Separating K groups whose memories lie within $\Delta\eta$ takes of order $(\Delta\eta)^{-(4K-2)}$ experiments; Prony's method attains this order, and passive observation needs at least as much data. The barrier does not lift as the market approaches instability: prices then reveal how close the market is to breaking, not who is trading. Table 1 gives the data this takes at illustrative parameters.

(4) *Supervision* (Section 5). A uniform cut in feedback activity needs only the kernel. Halting the largest group needs composition, at the cost above. Class-tagged flow reports remove the cost. A group that can mimic another group's memory caps what any price-based targeting can achieve.

(5) *Limits* (Section 6). With position limits the sufficient object becomes a measure over memory and leverage. Calm prices reveal only its linear part, the kernel. Probes large enough to reach the limits identify the rest.

What is known. Exponential memory is the oldest model of expectations: adaptive expectations (Cagan, 1956; Nerlove, 1958), geometric distributed lags (Koyck, 1954), and Muth (1960)'s proof that exponentially weighted forecasts are optimal for a random walk observed with noise. The idea that an aggregate depends on a distribution of individual coefficients and not on the individuals is also old. Granger (1980) showed it for sums of autoregressions, Forni and Lippi (1997) developed the general theory of aggregation and dynamics, Zaffaroni (2004) and Pesaran and Chudik (2014) the large-economy and large-panel versions, and recovering the distribution is the disaggregation problem (Celov et al., 2010). Results of the form "two structures give the same data, so data cannot choose" have a famous precedent in Sargent (1976), whose observational equivalence of natural and unnatural rate theories settled what reduced-form data could say about monetary rules. A large literature models markets with heterogeneous fundamentalist and chartist traders (Brock and Hommes, 1998; Hommes, 2006) and estimates their shares from prices (Boswijk et al., 2007; Franke and Westerhoff, 2012; Lux and Zwinkels, 2018); our results say which features of a population such estimates can recover from prices, and at what cost, when demand is linear in past prices. Auclert et al. (2024) make the corresponding sufficient-statistic argument for household heterogeneity in macroeconomics. The rate $(\Delta\eta)^{-(4K-2)}$ is the known minimax rate for mixtures with near-colliding components (Heinrich and Kahn, 2018; Wu and Yang, 2020), and the uniqueness step behind the stress result is the uniqueness of networks with one hidden layer (Sussmann, 1992). We do not claim these.

What is new. The market setting, and the policy question it answers: which supervisory actions the observational equivalence class supports, and which need data from outside it.

Sufficiency holds path by path inside a strategic market, for every insider and dealer rule, so it transfers to every equilibrium concept. The memory rates are transmission zeros of the closed-loop price response, not poles, which is why passive data are no cheaper than probing (Supplement Theorem S4.3). The results hold when price impact and persistence come from an inventory equilibrium (Supplement Theorem S3.2). Supervision splits into actions that need the kernel and actions that need composition, with a price on the second. With position limits, calm prices carry exactly no information about how close traders are to their limits.

2 Model

Time is discrete. Let x_t be the pricing error, the gap between the price and the fundamental value. A reduced-form dealer sector absorbs net order flow with price impact $\kappa > 0$, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$. There are K groups of feedback traders. Group i keeps an exponentially weighted average $y_{i,t}$ of past pricing errors and trades $q_i y_{i,t}$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + \varepsilon_t, \qquad y_{i,t} = (1 - \eta_i) y_{i,t-1} + \eta_i x_t, \tag{1}$$

where v_t is a probe, a trade by an observer, and ε_t is news. A desk that tracks a trend with a steady-state Kalman filter (Muth, 1960), or that learns by regularized online gradient descent, updates exactly this way (Supplement Lemma S1.1). Write $r_i = 1 - \eta_i$ for the memory rate and $w_i = q_i \eta_i$ for the trading weight. The *merged measure* $M = \sum_i w_i \delta_{r_i}$ puts weight w_i at rate r_i, merging groups with equal rates, and the *feedback kernel* is its sequence of moments,

$$k_m = \sum_i w_i r_i^m, \qquad m = 0, 1, 2, \ldots$$

Total feedback flow is the convolution of past prices with k. The *feedback budget* $S = a + \kappa \sum_i q_i = a + \kappa \sum_m k_m$ governs stability: with $q_i \geq 0$ the market is stable exactly when $S < 1$.

The observer. The cost results concern a supervisor or exchange that knows (a, κ) and the noise law. It can wait and record prices, probe from a reset state and record the response, or receive class-tagged flow reports. Reset probing is a benchmark for what data can reveal, not a proposed policy. Knowing (a, κ) only helps the observer, so the lower bounds below hold a fortiori when (a, κ) must be estimated.

3 The kernel is sufficient and is what prices identify

For sufficiency we embed the groups in a general market. A value v is drawn, noise traders submit z_t, an insider submits $X_t(v, \mathcal{H}_{t-1})$, and a dealer sets the price $P_t(\mathcal{H}_{t-1}, y_t)$ from the public history and the total order flow y_t. The groups update on a public signal $s_t = S_t(p_0, \ldots, p_t)$ and contribute flow $f_t = \sum_{m<t} k_m s_{t-1-m}$. The maps X, P and S are arbitrary causal rules, which covers Kyle, inventory-averse, learning and misspecified dealers.

Theorem 3.1 (The kernel is sufficient)**.** *If two group configurations have $k_m = k'_m$ for $m < T$, then for every strategy profile and every realization of (v, z) the price and order-flow paths, the insider's trades and realized profit, and the dealer's inventory coincide on dates $t \leq T$. The two games have the same strategy sets and payoffs, and hence the same equilibria under any concept defined from payoffs and the public history. The same holds for any finite population whose flows are causal linear functions of the public signal, with k the sum of the individual kernels. Group-level quantities, such as the weight of the largest group, are not invariant.*

The proof is an induction on dates (Supplement Theorem S2.1 and Proposition S2.2). It is short; its content is the object it isolates, a population-level quantity that settles every equilibrium concept at once. The induction uses only that aggregate feedback flow is a fixed causal function of the public signal. So for any population, linear or not, including traders who switch strategies after observing past profits, the aggregate flow map is sufficient (machine-checked in this generality; Supplement Theorem S6.1); the kernel is that map when flows are linear. For the same reason, observing aggregate order flow without class tags adds nothing about composition.

Theorem 3.2 (What prices identify)**.** *In model (1), two economies produce identical price responses to every probe sequence if and only if their merged measures coincide.*

This is Supplement Theorem S2.3(i). The same holds when κ and a are derived from competitive dealers who dislike inventory: the equilibrium price follows (1) path by path with effective impact and persistence, and with the memory rates unchanged (Supplement Theorems S3.1 and S3.2).

Together, Theorems 3.1 and 3.2 make the kernel a *minimal* sufficient statistic of the population for prices: it determines everything prices can show, and prices determine it. (The merged measure has finitely many atoms, so its moments, the kernel, determine it.)

Any population. For any population, linear or not, probe responses identify exactly the aggregate flow map, and a group-level quantity is identified if and only if it is a function of

that map (Supplement Theorem S6.7, machine-checked for an arbitrary flow map). Theorem 3.2 is the linear case. The lower bounds of the next section hold over any class of rules that contains the linear ones, so richer rules cannot make composition cheaper to learn.

4 The price of composition

Theorem 3.2 says that groups with different memories can be told apart in principle. The question is how much data that takes when memories are close.

Theorem 4.1 (The cost of composition)**.** *Fix $K \geq 2$, a center rate r_c, a total weight W and a spacing $\Delta\eta$, with every candidate economy stable. Each reset experiment records the noisy impulse response with Gaussian noise of variance σ^2.*

(i) Lower bound. *There are two K-group economies with rates in $r_c + \Delta\eta[-1, 1]$ and positive weights whose merged measures have the same moments of order $0, \dots, 2K-2$ and whose impulse responses differ by order $(\Delta\eta)^{2K-1}$. Telling them apart with error probability below $1/4$ needs $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments.*

(ii) Upper bound. *On any class of separated rescaled rates with weights bounded away from zero, Prony's method applied to the first $2K$ estimated kernel values recovers the weights from $N = C\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments.*

(iii) Waiting is no cheaper. *If prices are instead observed passively, driven by Gaussian news, the information per observation between the pair in (i) is of order $(\Delta\eta)^{4K-2}$, so at least of order $(\Delta\eta)^{-(4K-2)}$ observations are needed.*

Parts (i)–(iii) are Supplement Theorems S4.1, S4.2 and S4.4. Part (iii) does not weaken as the market approaches instability (Proposition 4.2). For two groups, halving the spread of memories multiplies the data needed by 64; for three groups, by 1,024. The pair in (i) consists of the Gauss and Gauss–Radau quadrature rules, which match moments as far as K nodes allow. Part (iii) uses that the memory rates are zeros of the closed-loop response, so passive data face the same near-cancellation as probes.

Proposition 4.2 (The barrier does not lift near instability)**.** *Hold the pair of Theorem 4.1(i) at a common feedback budget $S = 1 - \epsilon$. The passive information per observation between them is at most $C(\Delta\eta)^{4K-2}$, with C independent of ϵ, however close the market is to instability.*

This is Supplement Proposition S4.5. One might expect a market near instability to reveal its traders, since the loop amplifies every difference. It reveals instead how close it

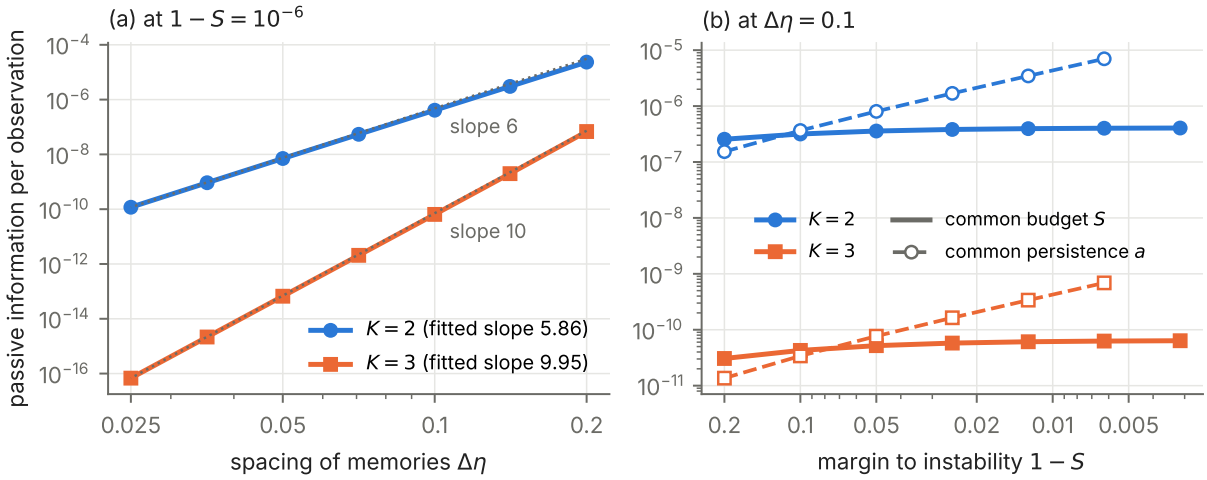


Figure 1: The passive barrier near instability. Both panels show the Gaussian Kullback–Leibler information per observation between the stationary price processes of two populations whose merged measures are the K-node Gauss and Gauss–Radau rules on memories spread over $\Delta\eta$ around 0.5 (total weight 0.3, impact $\kappa = 0.5$). (a) At a common feedback budget one millionth below instability, the information falls like $(\Delta\eta)^{4K-2}$ (dotted lines). (b) Held at a common budget (solid), the information stays bounded as the margin $1 - S$ shrinks; held at a common persistence (dashed), it grows without bound, because the budgets then differ. Computed from the closed-form spectra; scripts in `research_extension/fragility_reveals/`.

is to instability: with a common persistence rather than a common budget, the information grows without bound as $\epsilon \to 0$, and all of the growth comes from the budget. The bound is per observation; a finite-sample version uniform in ϵ is not proved, but in every case computed the exact information in n observations lies within 3% below n times the per-observation rate (Supplement Section S12.2). Figure 1 shows both statements.

How large is the barrier? Table 1 puts numbers on part (iii) for two groups. What matters is the spread of memories relative to their size. Populations whose memories all have half-lives between 5 and 9 days cannot be told apart with both error probabilities at 5% from less than 4,000 years of daily prices, however close the market is to instability. When half-lives differ by a factor of three or more, a century of daily data can suffice. With three groups the numbers grow to the order of 10^8 years. The numbers bound every test, for the passive record; probing faces the same order (part (i)).

Table 1: Years of daily price data below which no test tells the two populations of Theorem 4.1(i) apart with both error probabilities at 5%.

	memory rates	half-lives (days)	$1 - S = 0.1$	$1 - S = 10^{-6}$
$K = 2$	0.5 ± 0.1	0.8–1.4	20,000	16,000
	0.5 ± 0.05	0.9–1.2	1.2×10^6	9.0×10^5
	0.8 ± 0.1	1.9–6.6	100	69
	0.8 ± 0.05	2.4–4.3	6,000	3,900
	0.9 ± 0.05	4.3–13.5	120	77
	0.9 ± 0.025	5.2–8.9	7,000	4,300
$K = 3$	0.5 ± 0.1	0.8–1.4	1.5×10^8	9.9×10^7
	0.5 ± 0.05	0.9–1.2	1.5×10^{11}	9.6×10^{10}

Bound: $n \geq \ln 5/\mathrm{KL}$ observations (Bretagnolle–Huber), with KL the Gaussian information per observation, computed as in Figure 1, at a common feedback budget S, impact $\kappa = 0.5$ and feedback share $\kappa \sum_i q_i = 0.3$ (total weight $0.6(1 - r_c)$ at centre rate r_c). Memory rates are per day, 252 days a year; for traders whose memories are counted in five-minute periods, with prices recorded every five minutes around the clock, divide by 417. Two significant digits. Script: `research_extension/fragility_reveals/calibration_table.py`.

5 Who needs composition, and what disclosure buys

Proposition 5.1 (Market-wide versus targeted action)**.** *Let $q_i \geq 0$, let the market be stable with (a, κ) known, and let a supervisor need the feedback budget below a ceiling $\bar{s}$, with $a < \bar{s} < 1$.*

(i) Market-wide. *Scaling all feedback activity by θ meets the ceiling exactly when $a + \theta(S - a) \leq \bar{s}$. This depends on the population only through S, and so only through the kernel.*

(ii) Targeted. *Halting the largest group does not. For every $K \geq 2$ there are two populations whose merged measures have the same moments of order $0, \ldots, 2K - 2$, so that their budgets differ only by order $(\Delta\eta)^{2K-1}$, while their largest groups differ by a fixed amount, so that for a range of ceilings the halt meets the ceiling in one and not the other. Making this decision from N reset records with both errors at most $1/4$ needs $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)}$.*

Part (i) is Supplement Proposition S5.1(i). Part (ii) is Supplement Proposition S12.4; its pair comes from a family of quadrature rules whose weights are the values of one function at their nodes (Supplement Lemma S12.3).

Theorem 5.2 (Value of class-tagged disclosure)**.** *Suppose that, in addition to prices, the observer receives each class's flow with Gaussian measurement noise. On the rich class of Theorem 4.1(ii), the number of experiments needed to recover the weights falls from order $(\Delta\eta)^{-(4K-2)}$ to a bound that does not depend on $\Delta\eta$. For the rates it falls to order $(\Delta\eta)^{-2}$.*

This is Supplement Theorem S5.3; Supplement Proposition S5.2 gives the regression that achieves it, one class at a time, with a variance that depends only on that class.

The results so far take the groups' rules as given. A group that expects to be targeted has a reason to change them.

Proposition 5.3 (Hiding caps targeting)**.** *A supervisor observes prices, under any probes and for any record length, and then targets a given group or not. Being targeted costs the group $P > 0$. Suppose the group can adopt, at private cost C_0, a rule that gives the population the kernel of a population that the supervisor targets with probability α. Then in any best response the group is targeted with probability at most $\alpha + C_0/P$, whatever the probes, record length and decision rule.*

The proof is two lines from Theorem 3.1 (Supplement Proposition S5.4). More data make the group's original rule easier to detect and so make hiding more attractive. With several groups, the bound applies to each group's best response given the others' rules, so it holds group by group in any equilibrium; what determines each group's α is not modelled. A class tag is attached to the account, so changing the trading rule does not remove it.

6 Position limits: calm hides, stress reveals

Real feedback traders have position limits. Let trader i have memory rate r_i, capacity $c_i > 0$ and leverage ℓ_i, and trade $c_i \psi(\ell_i m_{i,t})$, where $m_{i,t}$ is its exponentially weighted signal and ψ is a shape common to all traders. The leading case is the hard limit $\psi(u) = \max(-1, \min(1, u))$: inside its band a trader is linear with weight $c_i \ell_i$; at the edge its position is capped. Write $\nu = \sum_i c_i \delta_{(r_i, \ell_i)}$ for the capacity measure on memory and leverage.

Theorem 6.1 (Two layers)**.** *(i)* Sufficiency. *For any shape, populations with the same ν give the same prices, profits and equilibria. The kernel is one linear functional of ν, $k_m = \int r^m \ell \, d\nu$, and does not determine it.*

(ii) Calm hides. *Under hard limits, two populations with the same kernel give identical price and order-flow paths until some trader first reaches its limit. Any record of the market before date T distinguishes them with total-variation distance at most the probability that a limit is reached before T. With Gaussian news and no probe this is at most $T(n + n')e^{-1/(2\vartheta^2)}$ for the $n + n'$ traders of the two populations, where ϑ is the largest standard deviation of a trader's position relative to its limit.*

(iii) Stress reveals. *If every trader has $r_i \geq 0$ and $\ell_i \neq 0$ and responses are observed exactly, probes of all sizes identify the signed capacity measure $\tilde{\mu} = \sum_i c_i \operatorname{sgn}(\ell_i) \, \delta_{(r_i, |\ell_i|)}$, and*

nothing finer, for the hard limit and for tanh*. For* tanh*, probes in any neighbourhood of zero suffice. For a cubic shape the result fails.*

Parts (i)–(iii) are Supplement Theorems S6.1, S6.2 and S6.4; for the hard limit and for tanh with probes of all sizes, part (iii) is machine-checked for any number of traders; the small-probe statement is proved on paper. The theorem assumes exactly observed responses; with observation noise, a minimum-distance estimator recovers the signed measure consistently, without a rate (Supplement Proposition S12.5). Its last step, for tanh, is the uniqueness of one-hidden-layer networks up to sign flips (Sussmann, 1992), which is why only the signed measure is identified. What the market adds is the reduction: probing recovers the flow map from prices, and two price dates give different memory–leverage pairs different scales.

The economic content is in part (ii). How close traders are to their limits, which governs behaviour in a crash, leaves no trace in calm prices. It can be learned only by pushing traders toward their limits or from disclosure.

7 Scope and conclusion

Evidence. The contribution is theoretical. Supplement Section S7 reports two pre-registered tests on public data, on Taiwan's tagged exchange flows and on Binance perpetual futures, with their readouts as registered.

The sufficiency results hold whenever aggregate feedback flow is a causal function of public signals; the kernel is the sufficient object when that function is linear, and ν when it saturates as in Section 6. The explicit identification and cost results need linear rules with exponential memory. For general rules, prices identify the aggregate flow map and the lower bounds still hold (Supplement Theorem S6.7), but no upper bound is proved, and for some families composition is not identified at all. For linear rules with general completely monotone memory the minimax accuracy is of order $1/\log(N/\sigma^2)$ (Niroula et al., 2026a). The passive results assume Gaussian news. Traders who change their rules as they learn are outside the theorems, apart from Proposition 5.3.

The problem this leaves. Together the results pose a problem. A supervisor who must keep a market stable cannot learn who trades in it from prices, at any reasonable cost, in calm conditions or near a crash; the traders can hide if it tries; and the information that would identify them arrives, if at all, only when limits are already being hit. Whether a market can be kept safe without that knowledge, using only what prices do reveal, is the question taken up in the companion paper (Niroula et al., 2026b).

What this says about the rules. The results sort the regulatory choices above. A market-wide measure, such as a uniform limit on feedback activity or a pause triggered by the feedback budget, needs no identification and can be set from prices. A measure aimed at particular algorithms cannot be set from prices at any reasonable cost, and the targeted algorithms can make it harder still by imitating others. It needs flow tagged by class or account, which is what the Chinese, Indian, Japanese and European regimes collect. Whether that data make the targeted measures worth their cost is an empirical question that only those tags can answer.

Within these limits the message is simple. Prices tell a supervisor the kernel and nothing more, and the kernel is enough for any action that treats the market as a whole. Actions aimed at one group need composition, which prices supply only at a steep and known price and which a group can hide at a bounded cost. Disclosure tagged by trader class removes that price. The same split recurs with position limits: calm markets reveal the kernel, and only stress or disclosure reveals how close traders are to their limits.

Declarations

Author contributions. Pratik Niroula: conceptualization, methodology, formal analysis, software, investigation, data curation, writing (original draft). Prashant Niroula: validation (verification of the proofs, the Lean 4 formalizations and the empirical calculations), writing (review and editing). Sujan Khatiwada: validation (verification of the proofs, the Lean 4 formalizations and the empirical calculations), writing (review and editing). All authors approved the final manuscript.

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Data availability. All data are public. The Taiwan test uses the Taiwan Stock Exchange's daily reports of trading by the three major institutional investors (T86), closing quotes (MI_INDEX), ex-rights and ex-dividend adjustments (TWT49U) and market totals (FMTQIK), from `https://www.twse.com.tw`. The Binance test uses daily files of five-minute candles and trading metrics (including open interest) for USD-margined perpetual futures from the Binance public data archive (`https://data.binance.vision`). The registered specifications, amendments, executors, notebooks, result files and test-unlock counters are in the public repository.

Code availability. All code, Lean 4 proofs and result files are public at https://github.com/Pratikn03/Fianince.

Ethics approval. Not applicable. The study uses only public market data and involves no human participants or personal data.

Declaration of generative AI and AI-assisted technologies. During the preparation of this work the authors used Claude, ChatGPT/Codex and Gemini for research development, code, numerical checks, Lean proofs, drafting and internal critique, as stated on the title page. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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